49. (**a**) For $z \neq 1$, verify the identity

$$1 + z + z^2 + \cdots + z^n = \frac{1 - z^{n+1}}{1 - z}.$$

(**b**) Use part (a) and appropriate results from this section to establish that

$$1 + \cos\theta + \cos 2\theta + \cdots + \cos n\theta = \frac{1}{2} + \frac{\sin\left(n + \frac{1}{2}\right)\theta}{2\sin\frac{1}{2}\theta}$$

for $0 < \theta < 2\pi$. The foregoing result is known as **Lagrange's identity** and is useful in the theory of Fourier series.

50. Suppose z_1, z_2, z_3, and z_4 are four distinct complex numbers. Interpret geometrically:

$$\arg\left(\frac{z_1 - z_2}{z_3 - z_4}\right) = \frac{\pi}{2}.$$

51. For n an even positive integer, establish the formula

$$\cos n\theta = \cos^n\theta - \binom{n}{2}\cos^{n-2}\theta\sin^2\theta + \binom{n}{4}\cos^{n-4}\theta\sin^4\theta + \cdots + (-1)^{n/2}\sin^n\theta$$

where $\binom{n}{k}$ is the **binomial coefficient** defined by

$$\binom{n}{k} = \frac{n(n-1)(n-2)\cdots(n-k+1)}{k!}.$$

Computer Lab Assignments

In Problems 52–55, use a CAS to find the modulus and principal argument of the given complex number. In *Mathematica* use the commands **Abs** and **Arg**.

52. $2 - 2i$

53. $\dfrac{-6i}{1 + \sqrt{3}i}$

54. $3.2 - 7.1i$

55. $\dfrac{(1.4 + 2.0i)(0.3 - 1.2i)}{6.5 - 3.9i}$

1.4 Powers and Roots

INTRODUCTION Recall from algebra that -2 and 2 are said to be *square roots* of the number 4 because $(-2)^2 = 4$ and $(2)^2 = 4$. In other words, the two square roots of 4 are distinct solutions of the equation $w^2 = 4$. In like manner we say $w = 3$ is a cube root of 27 since $w^3 = 3^3 = 27$. This last equation points us again in the direction of complex variables since any real number has only *one real* cube root and *two complex* roots. In general, we say that a number w is an ***n*th root** of a nonzero complex number z if $w^n = z$, where n is a positive integer. For example, you are urged to verify that $w_1 = \frac{1}{2}\sqrt{2} + \frac{1}{2}\sqrt{2}i$ and $w_2 = -\frac{1}{2}\sqrt{2} - \frac{1}{2}\sqrt{2}i$ are the two square roots of the complex number $z = i$ because $w_1^2 = i$ and $w_2^2 = i$. See Problem 39 in Exercises 1.1.

We will now demonstrate that there are exactly n solutions of the equation $w^n = z$.

Roots

Suppose $z = r(\cos\theta + i\sin\theta)$ and $w = \rho(\cos\phi + i\sin\phi)$ are polar forms of the complex numbers z and w. Then, in view of (9) of Section 1.3, the equation $w^n = z$ becomes

$$\rho^n(\cos n\phi + i\sin n\phi) = r(\cos\theta + i\sin\theta). \tag{1}$$

From (1), we can conclude that

$$\rho^n = r \tag{2}$$

and

$$\cos n\phi + i \sin n\phi = \cos\theta + i \sin\theta. \tag{3}$$

See Problem 47 in Exercises 1.3.

From (2), we define $\rho = \sqrt[n]{r}$ to be the unique positive nth root of the positive real number r. From (3), the definition of equality of two complex numbers implies that

$$\cos n\phi = \cos\theta \quad \text{and} \quad \sin n\phi = \sin\theta.$$

These equalities, in turn, indicate that the arguments θ and ϕ are related by $n\phi = \theta + 2k\pi$, where k is an integer. Thus,

$$\phi = \frac{\theta + 2k\pi}{n}.$$

As k takes on the successive integer values $k = 0,\ 1,\ 2,\ \ldots,\ n-1$ we obtain n *distinct* nth roots of z; these roots have the same modulus $\sqrt[n]{r}$ but different arguments. Notice that for $k \geq n$ we obtain the same roots because the sine and cosine are 2π-periodic. To see why this is so, suppose $k = n + m$, where $m = 0,\ 1,\ 2, \ldots$. Then

$$\phi = \frac{\theta + 2(n+m)\pi}{n} = \frac{\theta + 2m\pi}{n} + 2\pi$$

and

$$\sin\phi = \sin\left(\frac{\theta + 2m\pi}{n}\right), \quad \cos\phi = \cos\left(\frac{\theta + 2m\pi}{n}\right).$$

We summarize this result. The n nth roots of a nonzero complex number $z = r(\cos\theta + i\sin\theta)$ are given by

$$w_k = \sqrt[n]{r}\left[\cos\left(\frac{\theta + 2k\pi}{n}\right) + i\sin\left(\frac{\theta + 2k\pi}{n}\right)\right], \tag{4}$$

where $k = 0,\ 1,\ 2,\ \ldots,\ n-1$.

EXAMPLE 1 Cube Roots of a Complex Number

Find the three cube roots of $z = i$.

Solution Keep in mind that we are basically solving the equation $w^3 = i$. Now with $r = 1$, $\theta = \arg(i) = \pi/2$, a polar form of the given number is given by $z = \cos(\pi/2) + i\sin(\pi/2)$. From (4), with $n = 3$, we then obtain

$$w_k = \sqrt[3]{1}\left[\cos\left(\frac{\pi/2 + 2k\pi}{3}\right) + i\sin\left(\frac{\pi/2 + 2k\pi}{3}\right)\right], \quad k = 0, 1, 2.$$

Hence the three roots are,

$$k = 0, \quad w_0 = \cos\frac{\pi}{6} + i\sin\frac{\pi}{6} = \frac{\sqrt{3}}{2} + \frac{1}{2}i$$

$$k = 1, \quad w_1 = \cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6} = -\frac{\sqrt{3}}{2} + \frac{1}{2}i$$

$$k = 2, \quad w_2 = \cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2} = -i.$$

Principal *n*th Root

On page 16 we pointed out that the symbol $\arg(z)$ really stands for a *set* of arguments for a complex number z. Stated another way, for a given complex number $z \neq 0$, $\arg(z)$ is *infinite-valued*. In like manner, $z^{1/n}$ is n-valued; that is, the symbol $z^{1/n}$ represents the set of n nth roots w_k of z. The *unique* root of a complex number z (obtained by using the principal value of $\arg(z)$ with $k = 0$) is naturally referred to as the **principal *n*th root** of w. In Example 1, since $\text{Arg}(i) = \pi/2$, we see that $w_0 = \frac{1}{2}\sqrt{3} + \frac{1}{2}i$ is the principal cube root of i. The choice of $\theta = \text{Arg}(z)$ and $k = 0$ guarantees us that when z is a positive real number r, the principal nth root is $\sqrt[n]{r}$.

$\sqrt{4} = 2$ and $\sqrt[3]{27} = 3$ are the principal square root of 4 and the principal cube root of 27, respectively. ▶

Since the roots given by (4) have the same modulus, the n nth roots of a nonzero complex number z lie on a circle of radius $\sqrt[n]{r}$ centered at the origin in the complex plane. Moreover, since the difference between the arguments of any two successive roots w_k and w_{k+1} is $2\pi/n$, the n nth roots of z are equally spaced on this circle, beginning with the root whose argument is θ/n. FIGURE 1.4.1 shows the three cube roots of i obtained in Example 1 spaced at equal angular intervals of $2\pi/3$ on the circumference of a unit circle beginning with the root w_0 whose argument is $\pi/6$.

As the next example shows, the roots of a complex number do not have to be "nice" numbers as in Example 1.

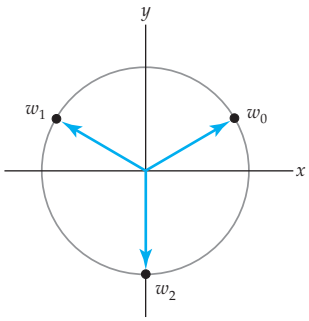


FIGURE 1.4.1 Three cube roots of i

EXAMPLE 2 Fourth Roots of a Complex Number

Find the four fourth roots of $z = 1 + i$.

Solution In this case, $r = \sqrt{2}$ and $\theta = \arg(z) = \pi/4$. From (4) with $n = 4$, we obtain

$$w_k = \sqrt[8]{2}\left[\cos\left(\frac{\pi/4 + 2k\pi}{4}\right) + i\sin\left(\frac{\pi/4 + 2k\pi}{4}\right)\right], \quad k = 0,\ 1,\ 2,\ 3.$$

With the aid of a calculator we find

$$\begin{aligned}
k = 0, \quad & w_0 = \sqrt[8]{2}\left[\cos\frac{\pi}{16} + i\sin\frac{\pi}{16}\right] \approx 1.0696 + 0.2127i \\
k = 1, \quad & w_1 = \sqrt[8]{2}\left[\cos\frac{9\pi}{16} + i\sin\frac{9\pi}{16}\right] \approx -0.2127 + 1.0696i \\
k = 2, \quad & w_2 = \sqrt[8]{2}\left[\cos\frac{17\pi}{16} + i\sin\frac{17\pi}{16}\right] \approx -1.0696 - 0.2127i \\
k = 3, \quad & w_3 = \sqrt[8]{2}\left[\cos\frac{25\pi}{16} + i\sin\frac{25\pi}{16}\right] \approx 0.2127 - 1.0696i.
\end{aligned}$$

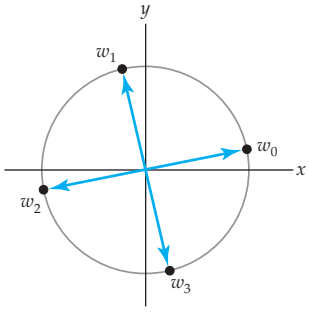


FIGURE 1.4.2 Four fourth roots of $1 + i$

As shown in FIGURE 1.4.2, the four roots lie on a circle centered at the origin of radius $r = \sqrt[8]{2} \approx 1.09$ and are spaced at equal angular intervals of $2\pi/4 = \pi/2$ radians, beginning with the root whose argument is $\pi/16$. Since $\theta = \text{Arg}(z) = \pi/4$, the principal fourth root is w_0. ☰

Remarks COMPARISON WITH REAL ANALYSIS

(*i*) As a consequence of (4), we can say that the complex number system is *closed* under the operation of extracting roots. This means that for any z in **C**, $z^{1/n}$ is also in **C**. The real number system does not possess a similar closure property since, if x is in **R**, $x^{1/n}$ is not necessarily in **R**.

(*ii*) Geometrically, the n nth roots of a complex number z can also be interpreted as the vertices of a regular polygon with n sides that is inscribed within a circle of radius $\sqrt[n]{r}$ centered at the origin. You can see the plausibility of this fact by reinspecting Figures 1.4.1 and 1.4.2. See Problem 19 in Exercises 1.4.

(*iii*) When m and n are positive integers with no common factors, then (4) enables us to define a **rational power** of z, that is, $z^{m/n}$. It can be shown that the set of values $(z^{1/n})^m$ is the same as the set of values $(z^m)^{1/n}$. This set of n common values is defined to be $z^{m/n}$. See Problems 25 and 26 in Exercises 1.4.

EXERCISES 1.4

Answers to selected odd-numbered problems begin on page ANS-2.

In Problems 1–14, use (4) to compute all roots. Give the principal nth root in each case. Sketch the roots $w_0, w_1, \ldots, w_{n-1}$ on an appropriate circle centered at the origin.

1. $(8)^{1/3}$

2. $(-1)^{1/4}$

3. $(-9)^{1/2}$

4. $(-125)^{1/3}$

5. $(i)^{1/2}$

6. $(-i)^{1/3}$

7. $(-1+i)^{1/3}$

8. $(1+i)^{1/5}$

9. $(-1+\sqrt{3}i)^{1/2}$

10. $(-1-\sqrt{3}i)^{1/4}$

11. $(3+4i)^{1/2}$

12. $(5+12i)^{1/2}$

13. $\left(\dfrac{16i}{1+i}\right)^{1/8}$

14. $\left(\dfrac{1+i}{\sqrt{3}+i}\right)^{1/6}$

15. (**a**) Verify that $(4+3i)^2 = 7+24i$.

(**b**) Use part (a) to find the two values of $(7+24i)^{1/2}$.

16. Rework Problem 15 using (4).

17. Find all solutions of the equation $z^4 + 1 = 0$.

18. Use the fact that $8i = (2+2i)^2$ to find all solutions of the equation

$$z^2 - 8z + 16 = 8i.$$

The n distinct ***n*th roots of unity** are the solutions of the equation $w^n = 1$. Problems 19–24 deal with roots of unity.

19. (**a**) Show that the n nth roots of unity are given by

$$(1)^{1/n} = \cos\frac{2k\pi}{n} + i\sin\frac{2k\pi}{n}, \quad k = 0, 1, 2, \ldots, n-1.$$

(**b**) Find n nth roots of unity for $n = 3$, $n = 4$, and $n = 5$.

(**c**) Carefully plot the roots of unity found in part (b). Sketch the regular polygons formed with the roots as vertices. [*Hint*: See (*ii*) in the *Remarks*.]

20. Suppose w is a cube root of unity corresponding to $k = 1$. See Problem 19(a).

(**a**) How are w and w^2 related?

(**b**) Verify by direct computation that

$$1 + w + w^2 = 0.$$

(**c**) Explain how the result in part (b) follows from the basic definition that w is a cube root of 1, that is, $w^3 = 1$. [*Hint*: Factor.]

21. For a fixed n, if we take $k = 1$ in Problem 19(a), we obtain the root

$$w_n = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}.$$

Explain why the n nth roots of unity can then be written

$$1, w_n, w_n^2, w_n^3, \ldots, w_n^{n-1}.$$

22. Consider the equation $(z + 2)^n + z^n = 0$, where n is a positive integer. By any means, solve the equation for z when $n = 1$. When $n = 2$.

23. Consider the equation in Problem 22.

(**a**) In the complex plane, determine the location of all solutions z when $n = 5$. [*Hint*: Write the equation in the form $[(z + 2)/(-z)]^5 = 1$ and use part (a) of Problem 19.]

(**b**) Reexamine the solutions of the equation in Problem 22 for $n = 1$ and $n = 2$. Form a conjecture as to the location of all solutions of $(z + 2)^n + z^n = 0$.

24. For the n nth roots of unity given in Problem 21, show that

$$1 + w_n + w_n^2 + w_n^3 + \cdots + w_n^{n-1} = 0.$$

[*Hint*: Multiply the sum $1 + w_n + w_n^2 + w_n^3 + \cdots + w_n^{n-1}$ by $w_n - 1$.]

Before working Problems 25 and 26, read (*iii*) in the *Remarks*. If m and n are positive integers with no common factors, then the n values of the rational power $z^{m/n}$ are

$$w_k = \sqrt[n]{r^m}\left[\cos\frac{m}{n}(\theta + 2k\pi) + i\sin\frac{m}{n}(\theta + 2k\pi)\right], \tag{5}$$

$k = 0,\ 1,\ 2,\ \ldots,\ n - 1$. The w_k are the n distinct solutions of $w^n = z^m$.

25. (**a**) First compute the set of values $i^{1/2}$ using (4). Then compute $(i^{1/2})^3$ using (9) of Section 1.3.

(**b**) Now compute i^3. Then compute $(i^3)^{1/2}$ using (4). Compare these values with the results of part (b).

(**c**) Lastly, compute $i^{3/2}$ using formula (5).

26. Use (5) to find all solutions of the equation $w^2 = (-1 + i)^5$.

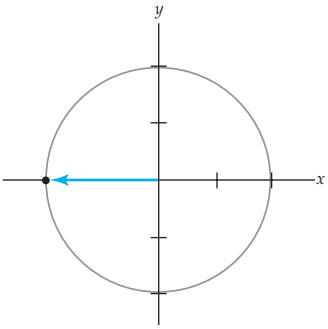


FIGURE 1.4.3 Figure for Problem 27

Focus on Concepts

27. The vector given in **FIGURE 1.4.3** represents one value of $z^{1/n}$. Using only the figure and trigonometry—that is, do not use formula (4)—find the remaining values of $z^{1/n}$ when $n = 3$. Write the values of $z^{1/n}$ in the form $a + ib$. Repeat for $n = 4$.